



BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 2 Note Sheets.

Bone Tissue · Axial Skeleton · Appendicular Skeleton · Joints

PART A

The brain dump prompts

All thirty Module 2 competencies with two drawing prompts each. Read this part once, then pick.

PART B

The note sheets

One sheet per competency. The competency and its two prompts on the left, one big drawing box on the right. Thirty sheets.

How this packet works

Every competency in Module 2 is on one sheet with two prompts beside one open box. You build one prompt in the box. The other is there if you want more practice, and it is a good place to go the week before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before any video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is the point. It is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing. Some want a grid. Some want three panels side by side. The division is yours to make.
4. Words are allowed inside the drawing, as labels and inside little boxes with arrows between them. A sentence running the width of the box is not. If a box reads like a paragraph, redo it as a picture.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a sorting grid or a sequence, so a student who thinks in pictures and a student who thinks in categories both get a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set the same thing up on a tablet, and copy the prompt letter you chose at the top. A hand ruled sheet counts exactly the same as a printed one.

This is anatomy

Every prompt asks what a structure is, where it sits, and what it is next to. None of them ask how it works. Function belongs to physiology and it is not on these sheets.

PART A

The brain dump prompts.

Thirty competencies, grouped by the concept video that teaches them. Each one gives you two prompts. Build one of them in that competency's box on your note sheet in Part B.

Bone tissue and microanatomy

1

Cartilage types and structure

WEEK 4

You should be able to: Identify the three cartilage types by fiber content and location and label chondrocytes, lacunae, matrix, and perichondrium.

LECTURE · LAB

A LABELED DRAWING, THREE PANELS

Divide the box into three panels, one per cartilage type. In each, draw the tissue as it looks under the microscope and label chondrocytes, lacunae, matrix, and the perichondrium where it exists. Draw the fibers so the three panels look different from each other.

Then: Under each panel, write one place in the body where you would find it.

B SORTING GRID

Draw a grid with hyaline, elastic, and fibrocartilage down the side. Across the top: the fiber that dominates, how the matrix looks on a slide, one location, and whether a perichondrium is present. Fill every cell with a word or a tiny sketch, not a sentence.

Then: Circle the one cell in the grid that lets you tell fibrocartilage from the other two at a glance.

2

Cartilage growth modes

WEEK 4

You should be able to: Distinguish appositional from interstitial cartilage growth by where each begins and the direction of expansion.

LECTURE

A SPLIT DRAWING

Draw a line down the middle. On the left, a piece of cartilage growing appositionally: show the perichondrium and arrows for where the new matrix is added. On the right, the same piece growing interstitially: show chondrocytes dividing inside the matrix and arrows for the direction it expands.

Then: Under each half, one line naming where that growth starts.

B BEFORE AND AFTER SEQUENCE

Draw a small block of cartilage three times in a row: before, during, and after growth. Do the row twice, once for appositional and once for interstitial, so the two rows sit one above the other and the difference in where the new tissue appears is obvious.

Then: Mark on each row the single structure that is the source of the new cartilage.

3

Bone classification by shape

WEEK 4

You should be able to: Classify a bone as long, short, flat, irregular, or sesamoid and give an example of each.

LECTURE

A SKELETON MAP

Draw a rough skeleton outline. Pick one bone for each of the five shapes, draw it in place, and label it with the shape it belongs to. Use a different mark for each shape so the five classes stand out.

Then: Add one more bone for the shape you find hardest to remember.

B FIVE-COLUMN SORT

Draw five columns headed long, short, flat, irregular, sesamoid. Under each heading, sketch the outline that defines the shape, then list two example bones. For sesamoid, draw the tendon the bone sits inside.

Then: Put a star beside the one example you think students most often put in the wrong column, and write which column they wrongly choose.

4

Gross anatomy of a long bone

WEEK 4

LECTURE

You should be able to: Label the diaphysis, epiphysis, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, epiphyseal line, and marrow regions on a long bone.

A LABELED SECTION

Draw a long bone cut down its length so the inside shows. Label diaphysis, both epiphyses, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, and the epiphyseal line. Shade compact bone and spongy bone differently and write where red and yellow marrow sit.

Then: Draw the nutrient foramen and the perforating fibers, and label both.

B OUTSIDE-IN LADDER

Draw a ladder that walks from the outside of a long bone to the center of its shaft: periosteum, compact bone, endosteum, medullary cavity, marrow. Then draw a second ladder for the end of the bone: articular cartilage, compact shell, spongy bone, red marrow. Beside each rung, a tiny sketch of what that layer looks like.

Then: Mark on the second ladder where the epiphyseal line sits and what it used to be.

5

Compact and spongy bone microanatomy

WEEK 4

LECTURE · LAB

You should be able to: Label the parts of an osteon and contrast compact bone with the trabeculae of spongy bone by structure and location.

A LABELED DRAWING, ONE OSTEON

Draw one osteon large in cross section. Central canal in the middle, concentric lamellae as rings, lacunae between the rings with an osteocyte in each, canaliculi as fine lines linking them. Add a perforating canal crossing to a neighboring osteon, and fit interstitial lamellae into the gaps between osteons.

Then: In the corner, draw a small patch of spongy bone and label a trabecula and the marrow space beside it.

B SIDE-BY-SIDE COMPARISON

Draw a line down the middle. Left: compact bone as it looks on a slide. Right: spongy bone on a slide. Under each, write its structural unit, what fills its spaces, and where in a long bone it sits.

Then: Circle the one structure that appears on the left and never on the right, and name it. That is your slide giveaway.

6

The four bone cells

WEEK 4

LECTURE · LAB

You should be able to: Identify the osteogenic cell, osteoblast, osteocyte, and osteoclast and state the role and origin of each.

A LINEAGE LADDER PLUS ONE OUTSIDER

Draw three rungs for the lineage, osteogenic cell to osteoblast to osteocyte, with a small drawing of each cell on its rung and where it sits in the bone. Draw the osteoclast off to the side, large and multinucleated, with its own arrow showing it comes from a blood cell line.

Then: Beside each cell write its job in three words or fewer.

B SORTING TABLE WITH SKETCHES

Draw a table with the four cells down the side. Across the top: what it looks like, where in the bone you find it, what it does, and where it comes from. Draw the cell in the first column instead of describing it.

Then: Put a box around the one row where the origin is different from the other three.

7

Ossification and growth plate zones

WEEK 4

LECTURE

You should be able to: Contrast intramembranous and endochondral ossification and list the five growth plate zones in order from epiphysis to diaphysis.

A LABELED GROWTH PLATE

Draw the growth plate as a tall column with the epiphysis at the top and the diaphysis at the bottom. Draw the five zones in order and label each: resting cartilage, proliferation, hypertrophy, calcification, ossification. Sketch what the chondrocytes look like in each zone so the zones are visibly different.

Then: Beside the column, write which zone lies against the epiphysis and which against the diaphysis.

B TWO-ROUTE SPLIT

Draw a line down the middle. Left: intramembranous ossification, the tissue it starts from, the bones it builds, and a small drawing of a skull bone. Right: endochondral ossification, the hyaline cartilage model it starts from, the bones it builds, and a small drawing of a long bone.

Then: Along the bottom, list the five growth plate zones in order as a single arrow from epiphysis to diaphysis.

The axial skeleton: skull, spine, and thoracic cage

8 Cranial and facial bones

WEEK 5

You should be able to: Distinguish the eight cranial bones from the fourteen facial bones and name the region each bone forms.

LECTURE · LAB

A LABELED SKULL, LATERAL VIEW

Draw a skull from the side. Label every cranial bone you can see and every facial bone you can see, and use two different marks, one for cranial and one for facial, so the two groups read at a glance. Write the count for each group.

Then: Draw a small anterior view beside it and add the facial bones that only show from the front.

B TWO-COLUMN SORT

Draw two columns headed cranial (8) and facial (14). List every bone, marking which are paired and which are single. Beside each bone, one phrase for the region it forms: the forehead, the cheek, the roof of the mouth, and so on.

Then: Circle the single unpaired bones in each column. There are fewer than you think.

9 Significant skull bone markings

WEEK 5

You should be able to: Identify the key markings on each cranial and facial bone, including features of the sphenoid, ethmoid, temporal, occipital, maxillae, and mandible.

LECTURE · LAB

A BONE BY BONE PANELS

Divide the box into six panels: sphenoid, ethmoid, temporal, occipital, maxilla, mandible. In each, draw the bone on its own, out of the skull, and label its key markings. The sphenoid and ethmoid panels need the view from above; the mandible needs the view from the side.

Then: In each panel, star the one marking you would use to identify that bone if it were handed to you loose.

B MARKING HUB MAP

Draw six hubs, one per bone, spread across the box. From each hub, spoke out to its markings with a tiny drawing of the marking at the end of each spoke: the sella turcica as a saddle, the mastoid process as a bump, the crista galli as a fin, and so on.

Then: Draw one line between two hubs for a marking that two bones share or meet at, and name it.

10 Sutures and fontanelles

WEEK 5

You should be able to: Locate the four major sutures and the pterion by the bones they join and name the four fontanelles and where each lies.

LECTURE · LAB

A TWO SKULLS, ADULT AND NEWBORN

Draw an adult skull from above and from the side. Draw and label the coronal, sagittal, lambdoid, and squamous sutures, and mark the pterion where the four bones meet. Then draw a newborn skull in the same two views and label the anterior, posterior, sphenoidal, and mastoid fontanelles.

Then: Draw an arrow from each fontanelle to the adult suture or junction it becomes.

B JOINING TABLE

Draw a table with the four sutures and the pterion down the side. Across the top: the bones it joins, where on the skull it runs, and a tiny sketch of its line. Below that, a second table for the four fontanelles: where each lies, which sutures meet there, and roughly when it closes.

Then: Circle the fontanelle a clinician checks on a baby and the junction a clinician worries about after a blow to the side of the head.

11 Orbit, nasal cavity, palate, and sinuses

WEEK 5

You should be able to: Identify the bones forming the orbit, nasal septum, and hard palate and locate the four paranasal sinuses by the bone each occupies.

LECTURE · LAB

A THREE LABELED DRAWINGS

Draw the orbit from the front and label every bone in its walls. Draw the nasal septum from the side and label the two bones and the cartilage that form it. Draw the hard palate from below and label the two bones that form it, front and back.

Then: On the orbit drawing, shade the four sinuses in the bones around it and name each.

B BONE-TO-SPACE GRID

Draw a grid with the bones down the side: frontal, ethmoid, sphenoid, maxilla, palatine, lacrimal, zygomatic, vomer, nasal. Across the top: orbit, nasal septum, hard palate, paranasal sinus. Tick every cell where that bone helps form that space.

Then: Beside the grid, draw a face outline and mark where each of the four sinuses sits.

12

Major skull foramina and contents

WEEK 5

You should be able to: Name the major skull foramina by the bone each pierces and state the nerve or vessel that passes through each.

LECTURE · LAB

A LABELED CRANIAL FLOOR

Draw the floor of the cranial cavity from above, the three cranial fossae as steps. Draw each major foramen as a hole in the right fossa, label it, and write beside it the bone it pierces and what passes through.

Then: Draw the foramen magnum last, larger than the rest, and label what passes through it.

B THREE-COLUMN TABLE

Draw a table with the foramina down the side: foramen magnum, optic canal, superior orbital fissure, foramen ovale, cribriform foramina, internal acoustic meatus, carotid canal, jugular foramen, mental foramen. Across the top: the bone, the fossa or region, and what passes through.

Then: Group the rows by bone with a bracket, so the sphenoid rows sit together.

13

Vertebral column regions and curvatures

WEEK 5

You should be able to: Name the five regions of the vertebral column with the vertebra count of each and identify the four curvatures as primary or secondary.

LECTURE

A LABELED SPINE, LATERAL VIEW

Draw the whole vertebral column from the side. Bracket the five regions and write the count in each. Draw the four curvatures clearly, and mark each one primary or secondary with a different mark.

Then: Beside the column, draw a tiny curled newborn spine to show which curves are there from the start.

B REGION AND CURVE TABLE

Draw a table with the five regions down the side and across the top: vertebra count, fused or separate, the curvature in that region, and whether that curve is primary or secondary. Draw the curve's shape in the cell instead of describing it.

Then: Add a row at the bottom for the total count of vertebrae in an adult and in a child, and show where the difference comes from.

14

Typical vertebra and intervertebral disc

WEEK 5

You should be able to: Label the parts of a typical vertebra and the anulus fibrosus and nucleus pulposus of an intervertebral disc.

LECTURE · LAB

A TWO LABELED VIEWS

Draw a typical vertebra from above and label the body, vertebral arch, pedicles, laminae, spinous process, transverse processes, vertebral foramen, and the superior articular processes. Then draw two vertebrae from the side with the disc between them and label the anulus fibrosus and nucleus pulposus.

Then: On the side view, mark the intervertebral foramen and label what passes through it.

B PART HUB

Draw a vertebra small in the middle of the box. Spoke out to every part and, at the end of each spoke, draw that part on its own and write what it is: a pedicle as a short bridge, a lamina as a plate, and so on. Add a second small hub for the disc with its two parts.

Then: Write beside the disc hub which of the two parts is the tough ring and which is the gel.

15

Regional and specialized vertebrae

WEEK 5

You should be able to: Distinguish cervical, thoracic, and lumbar vertebrae by an identifying feature and describe the atlas, axis, dens, sacrum, and coccyx.

LECTURE · LAB

A IDENTIFICATION PANELS

Draw a cervical, a thoracic, and a lumbar vertebra from above, side by side, and circle on each the feature that gives it away. Below them draw the atlas and the axis from above, label the dens, and draw the sacrum and coccyx from the front.

Then: Beside the atlas, write which bone sits on it and which bone the dens belongs to.

B GIVEAWAY TABLE

Draw a table with cervical, thoracic, lumbar, atlas, axis, sacrum, coccyx down the side. Across the top: the one feature that identifies it, the shape of its body, what its spinous process looks like, and a tiny sketch. Fill the sketch column with drawings, not words.

Then: Put a box around the two rows that have no body at all, and write why the atlas is missing one.

16

Sternum and ribs

WEEK 5

You should be able to: Identify the parts of the sternum and rib markings and classify the twelve rib pairs as true, false, or floating by anterior attachment.

LECTURE · LAB

A LABELED THORACIC CAGE

Draw the thoracic cage from the front. Label the manubrium, body, and xiphoid process of the sternum, and the sternal angle. Number the twelve ribs and use three marks to show true, false, and floating. Draw the costal cartilages so the anterior attachment of each rib is visible.

Then: Draw one rib on its own below and label the head, neck, tubercle, and shaft.

B ATTACHMENT SORT

Draw three columns headed true, false, floating. In each, list the rib numbers and draw how that group reaches, or fails to reach, the sternum. Below the columns, draw the sternum on its own and label its three parts and the sternal angle.

Then: Mark the rib that attaches at the sternal angle, and write why that landmark is useful for counting ribs.

CONCEPT VIDEO

The appendicular skeleton: limbs and girdles

17

Pectoral girdle and its markings

WEEK 6

You should be able to: Identify the markings of the clavicle and scapula and explain how the clavicle links the upper limb to the axial skeleton.

LECTURE · LAB

A LABELED BONES PLUS ONE LINK

Draw the scapula from behind and label the spine, acromion, coracoid process, glenoid cavity, supraspinous and infraspinous fossae, and the three borders and angles. Draw the clavicle above it and label the sternal and acromial ends. Then draw the sternum and show the one joint where the clavicle meets the axial skeleton.

Then: Draw a tiny chain: humerus, scapula, clavicle, sternum. That chain is the only bony route from arm to trunk.

B MARKING TABLE WITH SKETCHES

Draw a table with the scapula markings down the side and, below them, the clavicle markings. Across the top: where on the bone it is, what it looks like, and what meets it. Draw the marking in the second column.

Then: Add a final row named the link, and draw the sternoclavicular joint on it with the two bones labeled.

18

Humerus, radius, and ulna markings

WEEK 6

You should be able to: Identify the key markings of the humerus, radius, and ulna and state which forearm bone is lateral and which forms the elbow hinge.

LECTURE · LAB

A THREE LABELED BONES IN ANATOMICAL POSITION

Draw the humerus, radius, and ulna as they sit in anatomical position, palm forward, with the radius on the thumb side. Label the head, greater and lesser tubercles, deltoid tuberosity, epicondyles, trochlea, capitulum, and olecranon fossa on the humerus; the head, neck, radial tuberosity, and styloid process on the radius; the olecranon, trochlear notch, coronoid process, and styloid process on the ulna.

Then: Circle the two markings that lock together at the elbow hinge and draw an arrow between them.

B PROXIMAL-TO-DISTAL TABLE

Draw a table with three columns, humerus, radius, ulna, and rows that walk from proximal to distal. Put each marking in the row where it sits along the bone, so the table reads like the bones do. Draw the marking in its cell.

Then: Write across the bottom which bone is lateral and which bone forms the elbow hinge, then draw the tiny hinge.

19

Bones of the hand

WEEK 6

You should be able to: Name the eight carpals in their two rows and the metacarpals and phalanges and locate the carpal tunnel and its contents.

LECTURE · LAB

A LABELED HAND, PALM VIEW

Draw a hand from the palm side. Draw the eight carpals in two rows and label each one, lateral to medial. Number the metacarpals from the thumb, and draw the phalanges, showing that the thumb has two and every other finger has three.

Then: Draw the carpal tunnel as a cross section of the wrist below the hand: the carpal arch, the ligament roofing it, and the nerve and tendons inside.

B ROW AND COUNT GRID

Draw a grid. Row one: the four proximal carpals in order. Row two: the four distal carpals in order. Row three: the five metacarpals numbered. Row four: the phalanges, with the count for the thumb and for a finger. Beside each carpal, a tiny sketch of its shape.

Then: Add a box named carpal tunnel and list what forms its floor, what forms its roof, and what runs through it.

20

Bone markings vocabulary

WEEK 6

You should be able to: Classify a named bone marking as a projection, depression, or opening and match each term such as process, tuberosity, fossa, or sulcus to its description.

LECTURE

A THREE-FAMILY DRAWING SHEET

Divide the box into three bands: projections, depressions, openings. In each band, draw every term in that family as a tiny shape on a bone surface, a tubercle as a small bump, a tuberosity as a rough larger bump, a fossa as a shallow dish, a foramen as a hole, and label each shape with its term.

Then: Beside each shape, write one real bone marking that uses that term.

B SORTING TABLE

Draw a table with the terms down the side: process, tubercle, tuberosity, trochanter, crest, spine, condyle, epicondyle, head, facet, fossa, sulcus, foramen, meatus, fissure, sinus. Across the top: projection, depression, or opening; what it looks like; one example in the body.

Then: Mark the pairs of terms that students mix up most and draw the two side by side so the difference is visible.

21

Hip bone and pelvis

WEEK 6

You should be able to: Name the three bones that fuse at the acetabulum, identify the markings of the ilium, ischium, and pubis, and describe the pelvis and its sex differences.

LECTURE · LAB

A LABELED HIP BONE PLUS PELVIS

Draw one hip bone from the side. Shade the ilium, ischium, and pubis in three different ways and show them meeting at the acetabulum. Label the iliac crest, anterior superior iliac spine, greater sciatic notch, ischial tuberosity, ischial spine, obturator foramen, and pubic symphysis surface. Then draw the whole pelvis from the front, with the sacrum in place.

Then: Draw the female pelvis and the male pelvis side by side, small, and label the differences you can see: the angle under the pubis and the shape of the inlet.

B THREE-BONE TABLE AND A COMPARISON

Draw a table with ilium, ischium, pubis down the side. Across the top: where it sits in the hip bone, its markings, and what it contributes to the acetabulum and the obturator foramen. Below it, a second table with female and male across the top and the pelvic features that differ down the side.

Then: Draw the pelvic inlet for each sex as a small outline in the last row of the comparison table.

22

Femur and patella markings

WEEK 6

You should be able to: Identify the key markings of the femur and the base and apex of the patella.

LECTURE · LAB

A LABELED FEMUR, FRONT AND BACK

Draw the femur twice, from the front and from behind. Label the head, fovea, neck, greater and lesser trochanters, linea aspera, medial and lateral condyles and epicondyles, and the patellar surface. Draw the patella beside it and label the base and the apex.

Then: Mark which markings you can only see from behind, and draw the patella sitting on the femur so the base points the right way.

B PROXIMAL-TO-DISTAL LADDER

Draw the femur as a ladder from hip to knee. On each rung, draw and label the marking at that level, so the head and neck are at the top and the condyles at the bottom. Beside the ladder, draw the patella and label base and apex.

Then: Mark on the ladder which markings face the front, which face the back, and which face the sides.

23

Tibia and fibula markings

WEEK 6

LECTURE · LAB

You should be able to: Identify the markings of the tibia and fibula and state which bone bears weight and how to tell them apart.

A TWO LABELED BONES SIDE BY SIDE

Draw the tibia and fibula side by side as they sit in the leg, the fibula on the outside. Label the medial and lateral condyles, tibial tuberosity, anterior border, and medial malleolus on the tibia, and the head and lateral malleolus on the fibula. Draw them to scale so the difference in thickness is obvious.

Then: Draw a small cross section of the leg and mark which bone the weight travels through.

B TELL-THEM-APART TABLE

Draw a table with tibia and fibula across the top. Rows: which side of the leg, which bears weight, thick or thin, what it forms at the knee, what it forms at the ankle, and its markings. Draw the ankle malleoli in the ankle row so the medial and lateral bumps are labeled.

Then: In the last row, write the one feature you would use to tell a loose tibia from a loose fibula in lab.

24

Bones and arches of the foot

WEEK 6

LECTURE · LAB

You should be able to: Name the seven tarsals, the metatarsals, and the phalanges and identify the three arches of the foot.

A LABELED FOOT, TOP VIEW, PLUS SIDE VIEW

Draw the foot from above. Draw and label the seven tarsals, number the five metatarsals from the big toe, and draw the phalanges showing the big toe has two and the other toes have three. Then draw the foot from the inside and from the outside and draw the medial and lateral longitudinal arches. Draw a front-on cross section for the transverse arch.

Then: Circle the tarsal that carries the weight of the leg and the tarsal that forms the heel.

B BONE-TO-ARCH GRID

Draw a grid with the tarsals and metatarsals down the side. Across the top: proximal, intermediate, or distal group; the three arches. Tick every arch that bone helps form. Beside the grid, draw the two longitudinal arches as simple curves and the transverse arch as a dome.

Then: Beside the talus, draw the two leg bones that sit on top of it.

CONCEPT VIDEO

Articulations and joints

25

Structural and functional joint classification

WEEK 6

LECTURE

You should be able to: Classify a joint by structure as fibrous, cartilaginous, or synovial and by function as synarthrosis, amphiarthrosis, or diarthrosis and state the criteria for each.

A TWO-AXIS GRID

Draw a grid with the three structural classes down the side and the three functional classes across the top. In every cell that exists in the body, draw one example joint and name it. Leave the impossible cells empty and cross them out.

Then: Write beside the grid the one thing the structural classes are sorted by, and the one thing the functional classes are sorted by.

B TWO SORTING TREES

Draw two trees. Tree one starts with the question, what holds the bones together, and branches to fibrous, cartilaginous, synovial, each ending in a small drawing of that joint. Tree two starts with the question, how much does it move, and branches to synarthrosis, amphiarthrosis, diarthrosis, each ending in a body example.

Then: Draw a line between the two trees for every branch on one tree that always lands on the same branch of the other.

26

Fibrous and cartilaginous joints

WEEK 6

LECTURE

You should be able to: Describe the three fibrous joint types and the two cartilaginous joint types by structure, mobility, and example.

A FIVE LABELED DRAWINGS

Draw the five joints as simple cross sections: suture, syndesmosis, gomphosis, synchondrosis, symphysis. In each, draw what sits between the two bones and label it, dense fibrous tissue, a ligament, a periodontal ligament, hyaline cartilage, or fibrocartilage. Write the body example under each.

Then: Mark each drawing with how much it moves, and circle the one that is a tooth in its socket.

B SORTING TABLE

Draw a table with the five joint types down the side, grouped into fibrous and cartilaginous. Across the top: what connects the bones, how much it moves, its functional class, and one example. Draw the connecting tissue in the first column.

Then: Put a box around the two rows where the connecting tissue is cartilage and write which kind of cartilage each one uses.

27

Synovial joint structure and accessory parts

WEEK 6

LECTURE

You should be able to: Identify the structures of a synovial joint, its accessory structures such as menisci, labra, bursae, and tendon sheaths, and its nerve and blood supply.

A LABELED CROSS SECTION

Draw a synovial joint cut through the middle. Label the articular cartilage on each bone, the joint cavity, the synovial membrane, the fibrous layer of the capsule, and the synovial fluid. Add a bursa outside the capsule, a tendon sheath around a tendon, and a meniscus inside the cavity, and label all three.

Then: Draw a nerve and a small artery reaching the capsule, and write which layer each one supplies.

B INSIDE-OUT HUB

Put the joint cavity in the middle of the box. Spoke outward in layers: the fluid, the cartilage, the synovial membrane, the fibrous capsule, the ligaments. Then add a second ring of spokes for the accessory structures, meniscus, labrum, bursa, tendon sheath, each with a tiny drawing of where it sits.

Then: Write beside each accessory structure one joint in the body where you would find it.

28

Movements at synovial joints

WEEK 6

LECTURE

You should be able to: Demonstrate and name the gliding, angular, rotational, and special movements permitted at synovial joints.

A STICK FIGURE MOVEMENT SHEET

Draw a stick figure for every movement: flexion and extension, abduction and adduction, circumduction, rotation, pronation and supination, dorsiflexion and plantar flexion, inversion and eversion, protraction and retraction, elevation and depression, opposition. Show the starting position and the moved position with an arrow between them, and label each.

Then: Group the figures into gliding, angular, rotational, and special, and draw a box around each group.

B PAIRED-OPPOSITES TABLE

Draw a table with the movement pairs down the side and across the top: the joint where it happens, the plane it moves in, and a tiny stick-figure sketch. Fill the sketch column with drawings, and put the two members of each pair in one row so the opposites sit together.

Then: Circle the movements that have no opposite and write which joint each belongs to.

29

Six synovial joint types

WEEK 6

LECTURE

You should be able to: Name the six synovial joint types with their axes of movement and give a body example of each.

A SIX LABELED JOINT SHAPES

Draw the six joint types as the shapes of their bone ends: plane, hinge, pivot, condylar, saddle, ball-and-socket. Draw the arrows for the way each one moves, and write the body example under each. Draw them in order from the least movement to the most.

Then: Mark each shape with how many axes it moves around, and draw the axes as lines through the joint.

B AXES SORT

Draw three columns headed one axis, two axes, three axes. Put each of the six joint types in the right column with a small drawing of its shape and a body example. Add a fourth column for the one joint type that only glides.

Then: Beside each example joint, draw a tiny stick figure of the movement it allows.

You should be able to: Describe the articulating bones and supporting ligaments of the glenohumeral and tibiofemoral joints and identify the rotator cuff, menisci, and cruciate and collateral ligaments.

A TWO LABELED JOINTS

Draw the shoulder from the front: the glenoid cavity, the head of the humerus, the labrum, the capsule, and the four rotator cuff tendons wrapping the head. Then draw the knee from the front with the patella removed: the femoral condyles on the tibial plateau, both menisci, the two cruciate ligaments crossing, and the two collateral ligaments on the sides.

Then: On each joint, circle the structure that holds the joint together most, and write what the joint depends on when that structure is torn.

B TWO-JOINT COMPARISON TABLE

Draw a table with shoulder and knee across the top. Rows: the bones that meet, the shape of the joint, what deepens the socket, the ligaments that support it, the cartilage pads inside it, the muscles or tendons that stabilize it, and the direction it most often gives way. Draw the shape row instead of describing it.

Then: Draw a small side view of the knee under the table and label the cruciate ligaments so their crossing is visible.

PART B

The note sheets.

One sheet per competency, thirty sheets. The competency and both prompts sit on the left so nothing else has to be printed beside it. The box on the right is the whole assignment.

The box is **drawn**, not written. Pick one prompt and divide the box however it needs. Words go inside the drawing, in labels and little boxes with arrows between them. Sentences running across the page do not count.

1

COMP

WEEK 4 · BONE HISTOLOGY

Cartilage types and structure

You should be able to: Identify the three cartilage types by fiber content and location and label chondrocytes, lacunae, matrix, and perichondrium.

PICK ONE AND DO THAT ONE

☐ A

LABELLED DRAWING, THREE PANELS

Divide the box into three panels, one per cartilage type. In each, draw the tissue as it looks under the microscope and label chondrocytes, lacunae, matrix, and the perichondrium where it exists. Draw the fibers so the three panels look different from each other.

Then: Under each panel, write one place in the body where you would find it.

☐ B

SORTING GRID

Draw a grid with hyaline, elastic, and fibrocartilage down the side. Across the top: the fiber that dominates, how the matrix looks on a slide, one location, and whether a perichondrium is present. Fill every cell with a word or a tiny sketch, not a sentence.

Then: Circle the one cell in the grid that lets you tell fibrocartilage from the other two at a glance.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

DRAW IT

split it up however the prompt needs

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2

WEEK 4 · BONE HISTOLOGY

Cartilage growth modes

COMP

You should be able to: Distinguish appositional from interstitial cartilage growth by where each begins and the direction of expansion.

PICK ONE AND DO THAT ONE

☐ **A SPLIT DRAWING**

Draw a line down the middle. On the left, a piece of cartilage growing appositionally: show the perichondrium and arrows for where the new matrix is added. On the right, the same piece growing interstitially: show chondrocytes dividing inside the matrix and arrows for the direction it expands.

Then: Under each half, one line naming where that growth starts.

☐ **B BEFORE AND AFTER SEQUENCE**

Draw a small block of cartilage three times in a row: before, during, and after growth. Do the row twice, once for appositional and once for interstitial, so the two rows sit one above the other and the difference in where the new tissue appears is obvious.

Then: Mark on each row the single structure that is the source of the new cartilage.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

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3

COMP

WEEK 4 · BONE HISTOLOGY

Bone classification by shape

You should be able to: Classify a bone as long, short, flat, irregular, or sesamoid and give an example of each.

PICK ONE AND DO THAT ONE

- ☐ **A SKELETON MAP**
Draw a rough skeleton outline. Pick one bone for each of the five shapes, draw it in place, and label it with the shape it belongs to. Use a different mark for each shape so the five classes stand out.

Then: Add one more bone for the shape you find hardest to remember.
- ☐ **B FIVE-COLUMN SORT**
Draw five columns headed long, short, flat, irregular, sesamoid. Under each heading, sketch the outline that defines the shape, then list two example bones. For sesamoid, draw the tendon the bone sits inside.

Then: Put a star beside the one example you think students most often put in the wrong column, and write which column they wrongly choose.

PASS 1 COLOR _____
PASS 2 COLOR _____

DRAW IT

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4

COMP

WEEK 4 · BONE HISTOLOGY

Gross anatomy of a long bone

You should be able to: Label the diaphysis, epiphysis, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, epiphyseal line, and marrow regions on a long bone.

PICK ONE AND DO THAT ONE

☐ **A LABELED SECTION**

Draw a long bone cut down its length so the inside shows. Label diaphysis, both epiphyses, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, and the epiphyseal line. Shade compact bone and spongy bone differently and write where red and yellow marrow sit.

Then: Draw the nutrient foramen and the perforating fibers, and label both.

☐ **B OUTSIDE-IN LADDER**

Draw a ladder that walks from the outside of a long bone to the center of its shaft: periosteum, compact bone, endosteum, medullary cavity, marrow. Then draw a second ladder for the end of the bone: articular cartilage, compact shell, spongy bone, red marrow. Beside each rung, a tiny sketch of what that layer looks like.

Then: Mark on the second ladder where the epiphyseal line sits and what it used to be.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

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5

COMP

WEEK 4 · BONE HISTOLOGY

Compact and spongy bone microanatomy

You should be able to: Label the parts of an osteon and contrast compact bone with the trabeculae of spongy bone by structure and location.

PICK ONE AND DO THAT ONE

☐ **A LABELED DRAWING, ONE OSTEON**

Draw one osteon large in cross section. Central canal in the middle, concentric lamellae as rings, lacunae between the rings with an osteocyte in each, canaliculi as fine lines linking them. Add a perforating canal crossing to a neighboring osteon, and fit interstitial lamellae into the gaps between osteons.

Then: In the corner, draw a small patch of spongy bone and label a trabecula and the marrow space beside it.

☐ **B SIDE-BY-SIDE COMPARISON**

Draw a line down the middle. Left: compact bone as it looks on a slide. Right: spongy bone on a slide. Under each, write its structural unit, what fills its spaces, and where in a long bone it sits.

Then: Circle the one structure that appears on the left and never on the right, and name it. That is your slide giveaway.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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6

COMP

WEEK 4 · BONE HISTOLOGY

The four bone cells

You should be able to: Identify the osteogenic cell, osteoblast, osteocyte, and osteoclast and state the role and origin of each.

PICK ONE AND DO THAT ONE

- ☐ **A LINEAGE LADDER PLUS ONE OUTSIDER**
Draw three rungs for the lineage, osteogenic cell to osteoblast to osteocyte, with a small drawing of each cell on its rung and where it sits in the bone. Draw the osteoclast off to the side, large and multinucleated, with its own arrow showing it comes from a blood cell line.

Then: Beside each cell write its job in three words or fewer.
- ☐ **B SORTING TABLE WITH SKETCHES**
Draw a table with the four cells down the side. Across the top: what it looks like, where in the bone you find it, what it does, and where it comes from. Draw the cell in the first column instead of describing it.

Then: Put a box around the one row where the origin is different from the other three.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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7

COMP

WEEK 4 · BONE HISTOLOGY

Ossification and growth plate zones

You should be able to: Contrast intramembranous and endochondral ossification and list the five growth plate zones in order from epiphysis to diaphysis.

PICK ONE AND DO THAT ONE

☐ **A LABELED GROWTH PLATE**

Draw the growth plate as a tall column with the epiphysis at the top and the diaphysis at the bottom. Draw the five zones in order and label each: resting cartilage, proliferation, hypertrophy, calcification, ossification. Sketch what the chondrocytes look like in each zone so the zones are visibly different.

Then: Beside the column, write which zone lies against the epiphysis and which against the diaphysis.

☐ **B TWO-ROUTE SPLIT**

Draw a line down the middle. Left: intramembranous ossification, the tissue it starts from, the bones it builds, and a small drawing of a skull bone. Right: endochondral ossification, the hyaline cartilage model it starts from, the bones it builds, and a small drawing of a long bone.

Then: Along the bottom, list the five growth plate zones in order as a single arrow from epiphysis to diaphysis.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

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8

COMP

WEEK 5 · AXIAL SKELETON, SKULL

Cranial and facial bones

You should be able to: Distinguish the eight cranial bones from the fourteen facial bones and name the region each bone forms.

PICK ONE AND DO THAT ONE

- ☐ A

LABELED SKULL, LATERAL VIEW

Draw a skull from the side. Label every cranial bone you can see and every facial bone you can see, and use two different marks, one for cranial and one for facial, so the two groups read at a glance. Write the count for each group.

Then: Draw a small anterior view beside it and add the facial bones that only show from the front.
- ☐ B

TWO-COLUMN SORT

Draw two columns headed cranial (8) and facial (14). List every bone, marking which are paired and which are single. Beside each bone, one phrase for the region it forms: the forehead, the cheek, the roof of the mouth, and so on.

Then: Circle the single unpaired bones in each column. There are fewer than you think.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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9

COMP

WEEK 5 · AXIAL SKELETON, SKULL

Significant skull bone markings

You should be able to: Identify the key markings on each cranial and facial bone, including features of the sphenoid, ethmoid, temporal, occipital, maxillae, and mandible.

PICK ONE AND DO THAT ONE

- ☐ **A BONE BY BONE PANELS**
Divide the box into six panels: sphenoid, ethmoid, temporal, occipital, maxilla, mandible. In each, draw the bone on its own, out of the skull, and label its key markings. The sphenoid and ethmoid panels need the view from above; the mandible needs the view from the side.

Then: In each panel, star the one marking you would use to identify that bone if it were handed to you loose.
- ☐ **B MARKING HUB MAP**
Draw six hubs, one per bone, spread across the box. From each hub, spoke out to its markings with a tiny drawing of the marking at the end of each spoke: the sella turcica as a saddle, the mastoid process as a bump, the crista galli as a fin, and so on.

Then: Draw one line between two hubs for a marking that two bones share or meet at, and name it.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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10

WEEK 5 · AXIAL SKELETON, SKULL

Sutures and fontanelles

COMP

You should be able to: Locate the four major sutures and the pterion by the bones they join and name the four fontanelles and where each lies.

PICK ONE AND DO THAT ONE

☐ **A TWO SKULLS, ADULT AND NEWBORN**

Draw an adult skull from above and from the side. Draw and label the coronal, sagittal, lambdoid, and squamous sutures, and mark the pterion where the four bones meet. Then draw a newborn skull in the same two views and label the anterior, posterior, sphenoidal, and mastoid fontanelles.

Then: Draw an arrow from each fontanelle to the adult suture or junction it becomes.

☐ **B JOINING TABLE**

Draw a table with the four sutures and the pterion down the side. Across the top: the bones it joins, where on the skull it runs, and a tiny sketch of its line. Below that, a second table for the four fontanelles: where each lies, which sutures meet there, and roughly when it closes.

Then: Circle the fontanelle a clinician checks on a baby and the junction a clinician worries about after a blow to the side of the head.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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11

COMP

WEEK 5 · AXIAL SKELETON, SKULL

Orbit, nasal cavity, palate, and sinuses

You should be able to: Identify the bones forming the orbit, nasal septum, and hard palate and locate the four paranasal sinuses by the bone each occupies.

PICK ONE AND DO THAT ONE

☐ **A THREE LABELED DRAWINGS**

Draw the orbit from the front and label every bone in its walls. Draw the nasal septum from the side and label the two bones and the cartilage that form it. Draw the hard palate from below and label the two bones that form it, front and back.

Then: On the orbit drawing, shade the four sinuses in the bones around it and name each.

☐ **B BONE-TO-SPACE GRID**

Draw a grid with the bones down the side: frontal, ethmoid, sphenoid, maxilla, palatine, lacrimal, zygomatic, vomer, nasal. Across the top: orbit, nasal septum, hard palate, paranasal sinus. Tick every cell where that bone helps form that space.

Then: Beside the grid, draw a face outline and mark where each of the four sinuses sits.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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12

COMP

WEEK 5 · AXIAL SKELETON, SKULL

Major skull foramina and contents

You should be able to: Name the major skull foramina by the bone each pierces and state the nerve or vessel that passes through each.

PICK ONE AND DO THAT ONE

☐ **A LABELED CRANIAL FLOOR**

Draw the floor of the cranial cavity from above, the three cranial fossae as steps. Draw each major foramen as a hole in the right fossa, label it, and write beside it the bone it pierces and what passes through.

Then: Draw the foramen magnum last, larger than the rest, and label what passes through it.

☐ **B THREE-COLUMN TABLE**

Draw a table with the foramina down the side: foramen magnum, optic canal, superior orbital fissure, foramen ovale, cribriform foramina, internal acoustic meatus, carotid canal, jugular foramen, mental foramen. Across the top: the bone, the fossa or region, and what passes through.

Then: Group the rows by bone with a bracket, so the sphenoid rows sit together.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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13

COMP

WEEK 5 · AXIAL SKELETON, SPINE

Vertebral column regions and curvatures

You should be able to: Name the five regions of the vertebral column with the vertebra count of each and identify the four curvatures as primary or secondary.

PICK ONE AND DO THAT ONE

- ☐ **A** LABELED SPINE, LATERAL VIEW
- Draw the whole vertebral column from the side. Bracket the five regions and write the count in each. Draw the four curvatures clearly, and mark each one primary or secondary with a different mark.
- Then: Beside the column, draw a tiny curled newborn spine to show which curves are there from the start.
- ☐ **B** REGION AND CURVE TABLE
- Draw a table with the five regions down the side and across the top: vertebra count, fused or separate, the curvature in that region, and whether that curve is primary or secondary. Draw the curve's shape in the cell instead of describing it.
- Then: Add a row at the bottom for the total count of vertebrae in an adult and in a child, and show where the difference comes from.

PASS 1 COLOR _____

PASS 2 COLOR _____

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14

COMP

WEEK 5 · AXIAL SKELETON, SPINE

Typical vertebra and intervertebral disc

You should be able to: Label the parts of a typical vertebra and the anulus fibrosus and nucleus pulposus of an intervertebral disc.

PICK ONE AND DO THAT ONE

☐ **A TWO LABELED VIEWS**

Draw a typical vertebra from above and label the body, vertebral arch, pedicles, laminae, spinous process, transverse processes, vertebral foramen, and the superior articular processes. Then draw two vertebrae from the side with the disc between them and label the anulus fibrosus and nucleus pulposus.

Then: On the side view, mark the intervertebral foramen and label what passes through it.

☐ **B PART HUB**

Draw a vertebra small in the middle of the box. Spoke out to every part and, at the end of each spoke, draw that part on its own and write what it is: a pedicle as a short bridge, a lamina as a plate, and so on. Add a second small hub for the disc with its two parts.

Then: Write beside the disc hub which of the two parts is the tough ring and which is the gel.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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15

COMP

WEEK 5 · AXIAL SKELETON, SPINE

Regional and specialized vertebrae

You should be able to: Distinguish cervical, thoracic, and lumbar vertebrae by an identifying feature and describe the atlas, axis, dens, sacrum, and coccyx.

PICK ONE AND DO THAT ONE

- ☐ **A IDENTIFICATION PANELS**
Draw a cervical, a thoracic, and a lumbar vertebra from above, side by side, and circle on each the feature that gives it away. Below them draw the atlas and the axis from above, label the dens, and draw the sacrum and coccyx from the front.

Then: Beside the atlas, write which bone sits on it and which bone the dens belongs to.
- ☐ **B GIVEAWAY TABLE**
Draw a table with cervical, thoracic, lumbar, atlas, axis, sacrum, coccyx down the side. Across the top: the one feature that identifies it, the shape of its body, what its spinous process looks like, and a tiny sketch. Fill the sketch column with drawings, not words.

Then: Put a box around the two rows that have no body at all, and write why the atlas is missing one.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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16

WEEK 5 · AXIAL SKELETON, SPINE

Sternum and ribs

COMP

You should be able to: Identify the parts of the sternum and rib markings and classify the twelve rib pairs as true, false, or floating by anterior attachment.

PICK ONE AND DO THAT ONE

☐ **A LABELED THORACIC CAGE**

Draw the thoracic cage from the front. Label the manubrium, body, and xiphoid process of the sternum, and the sternal angle. Number the twelve ribs and use three marks to show true, false, and floating. Draw the costal cartilages so the anterior attachment of each rib is visible.

Then: Draw one rib on its own below and label the head, neck, tubercle, and shaft.

☐ **B ATTACHMENT SORT**

Draw three columns headed true, false, floating. In each, list the rib numbers and draw how that group reaches, or fails to reach, the sternum. Below the columns, draw the sternum on its own and label its three parts and the sternal angle.

Then: Mark the rib that attaches at the sternal angle, and write why that landmark is useful for counting ribs.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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17

COMP

WEEK 6 · APPENDICULAR, UPPER EXTREMITY

Pectoral girdle and its markings

You should be able to: Identify the markings of the clavicle and scapula and explain how the clavicle links the upper limb to the axial skeleton.

PICK ONE AND DO THAT ONE

- ☐ **A** **LABELED BONES PLUS ONE LINK**
Draw the scapula from behind and label the spine, acromion, coracoid process, glenoid cavity, supraspinous and infraspinous fossae, and the three borders and angles. Draw the clavicle above it and label the sternal and acromial ends. Then draw the sternum and show the one joint where the clavicle meets the axial skeleton.

Then: Draw a tiny chain: humerus, scapula, clavicle, sternum. That chain is the only bony route from arm to trunk.
- ☐ **B** **MARKING TABLE WITH SKETCHES**
Draw a table with the scapula markings down the side and, below them, the clavicle markings. Across the top: where on the bone it is, what it looks like, and what meets it. Draw the marking in the second column.

Then: Add a final row named the link, and draw the sternoclavicular joint on it with the two bones labeled.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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18

COMP

WEEK 6 · APPENDICULAR, UPPER EXTREMITY

Humerus, radius, and ulna markings

You should be able to: Identify the key markings of the humerus, radius, and ulna and state which forearm bone is lateral and which forms the elbow hinge.

PICK ONE AND DO THAT ONE

- ☐ **A THREE LABELED BONES IN ANATOMICAL POSITION**
Draw the humerus, radius, and ulna as they sit in anatomical position, palm forward, with the radius on the thumb side. Label the head, greater and lesser tubercles, deltoid tuberosity, epicondyles, trochlea, capitulum, and olecranon fossa on the humerus; the head, neck, radial tuberosity, and styloid process on the radius; the olecranon, trochlear notch, coronoid process, and styloid process on the ulna.

Then: Circle the two markings that lock together at the elbow hinge and draw an arrow between them.
- ☐ **B PROXIMAL-TO-DISTAL TABLE**
Draw a table with three columns, humerus, radius, ulna, and rows that walk from proximal to distal. Put each marking in the row where it sits along the bone, so the table reads like the bones do. Draw the marking in its cell.

Then: Write across the bottom which bone is lateral and which bone forms the elbow hinge, then draw the tiny hinge.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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19

WEEK 6 · APPENDICULAR, UPPER EXTREMITY

COMP

Bones of the hand

You should be able to: Name the eight carpals in their two rows and the metacarpals and phalanges and locate the carpal tunnel and its contents.

PICK ONE AND DO THAT ONE

☐ **A LABELED HAND, PALM VIEW**

Draw a hand from the palm side. Draw the eight carpals in two rows and label each one, lateral to medial. Number the metacarpals from the thumb, and draw the phalanges, showing that the thumb has two and every other finger has three.

Then: Draw the carpal tunnel as a cross section of the wrist below the hand: the carpal arch, the ligament roofing it, and the nerve and tendons inside.

☐ **B ROW AND COUNT GRID**

Draw a grid. Row one: the four proximal carpals in order. Row two: the four distal carpals in order. Row three: the five metacarpals numbered. Row four: the phalanges, with the count for the thumb and for a finger. Beside each carpal, a tiny sketch of its shape.

Then: Add a box named carpal tunnel and list what forms its floor, what forms its roof, and what runs through it.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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20

COMP

WEEK 6 · APPENDICULAR SKELETON

Bone markings vocabulary

You should be able to:

Classify a named bone marking as a projection, depression, or opening and match each term such as process, tuberosity, fossa, or sulcus to its description.

PICK ONE AND DO THAT ONE

- ☐ A

THREE-FAMILY DRAWING SHEET

Divide the box into three bands: projections, depressions, openings. In each band, draw every term in that family as a tiny shape on a bone surface, a tubercle as a small bump, a tuberosity as a rough larger bump, a fossa as a shallow dish, a foramen as a hole, and label each shape with its term.

Then: Beside each shape, write one real bone marking that uses that term.
- ☐ B

SORTING TABLE

Draw a table with the terms down the side: process, tubercle, tuberosity, trochanter, crest, spine, condyle, epicondyle, head, facet, fossa, sulcus, foramen, meatus, fissure, sinus. Across the top: projection, depression, or opening; what it looks like; one example in the body.

Then: Mark the pairs of terms that students mix up most and draw the two side by side so the difference is visible.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

split it up however the prompt needs

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21

COMP

WEEK 6 · APPENDICULAR, LOWER EXTREMITY

Hip bone and pelvis

You should be able to: Name the three bones that fuse at the acetabulum, identify the markings of the ilium, ischium, and pubis, and describe the pelvis and its sex differences.

PICK ONE AND DO THAT ONE

- ☐ **A LABELED HIP BONE PLUS PELVIS**
Draw one hip bone from the side. Shade the ilium, ischium, and pubis in three different ways and show them meeting at the acetabulum. Label the iliac crest, anterior superior iliac spine, greater sciatic notch, ischial tuberosity, ischial spine, obturator foramen, and pubic symphysis surface. Then draw the whole pelvis from the front, with the sacrum in place.

Then: Draw the female pelvis and the male pelvis side by side, small, and label the differences you can see: the angle under the pubis and the shape of the inlet.
- ☐ **B THREE-BONE TABLE AND A COMPARISON**
Draw a table with ilium, ischium, pubis down the side. Across the top: where it sits in the hip bone, its markings, and what it contributes to the acetabulum and the obturator foramen. Below it, a second table with female and male across the top and the pelvic features that differ down the side.

Then: Draw the pelvic inlet for each sex as a small outline in the last row of the comparison table.

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PASS 2 COLOR _____

ALSO A LAB SKILL

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WEEK 6 · APPENDICULAR, LOWER EXTREMITY

COMP

Femur and patella markings

You should be able to: Identify the key markings of the femur and the base and apex of the patella.

PICK ONE AND DO THAT ONE

☐ **A LABELED FEMUR, FRONT AND BACK**

Draw the femur twice, from the front and from behind. Label the head, fovea, neck, greater and lesser trochanters, linea aspera, medial and lateral condyles and epicondyles, and the patellar surface. Draw the patella beside it and label the base and the apex.

Then: Mark which markings you can only see from behind, and draw the patella sitting on the femur so the base points the right way.

☐ **B PROXIMAL-TO-DISTAL LADDER**

Draw the femur as a ladder from hip to knee. On each rung, draw and label the marking at that level, so the head and neck are at the top and the condyles at the bottom. Beside the ladder, draw the patella and label base and apex.

Then: Mark on the ladder which markings face the front, which face the back, and which face the sides.

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ALSO A LAB SKILL

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COMP

WEEK 6 · APPENDICULAR, LOWER EXTREMITY

Tibia and fibula markings

You should be able to: Identify the markings of the tibia and fibula and state which bone bears weight and how to tell them apart.

PICK ONE AND DO THAT ONE

- ☐ **A TWO LABELED BONES SIDE BY SIDE**
Draw the tibia and fibula side by side as they sit in the leg, the fibula on the outside. Label the medial and lateral condyles, tibial tuberosity, anterior border, and medial malleolus on the tibia, and the head and lateral malleolus on the fibula. Draw them to scale so the difference in thickness is obvious.

Then: Draw a small cross section of the leg and mark which bone the weight travels through.
- ☐ **B TELL-THEM-APART TABLE**
Draw a table with tibia and fibula across the top. Rows: which side of the leg, which bears weight, thick or thin, what it forms at the knee, what it forms at the ankle, and its markings. Draw the ankle malleoli in the ankle row so the medial and lateral bumps are labeled.

Then: In the last row, write the one feature you would use to tell a loose tibia from a loose fibula in lab.

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WEEK 6 · APPENDICULAR, LOWER EXTREMITY

COMP

Bones and arches of the foot

You should be able to: Name the seven tarsals, the metatarsals, and the phalanges and identify the three arches of the foot.

PICK ONE AND DO THAT ONE

☐ **A LABELED FOOT, TOP VIEW, PLUS SIDE VIEW**

Draw the foot from above. Draw and label the seven tarsals, number the five metatarsals from the big toe, and draw the phalanges showing the big toe has two and the other toes have three. Then draw the foot from the inside and from the outside and draw the medial and lateral longitudinal arches. Draw a front-on cross section for the transverse arch.

Then: Circle the tarsal that carries the weight of the leg and the tarsal that forms the heel.

☐ **B BONE-TO-ARCH GRID**

Draw a grid with the tarsals and metatarsals down the side. Across the top: proximal, intermediate, or distal group; the three arches. Tick every arch that bone helps form. Beside the grid, draw the two longitudinal arches as simple curves and the transverse arch as a dome.

Then: Beside the talus, draw the two leg bones that sit on top of it.

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COMP

WEEK 6 · ARTICULATIONS AND JOINTS

Structural and functional joint classification

You should be able to: Classify a joint by structure as fibrous, cartilaginous, or synovial and by function as synarthrosis, amphiarthrosis, or diarthrosis and state the criteria for each.

PICK ONE AND DO THAT ONE

- ☐ **A TWO-AXIS GRID**
Draw a grid with the three structural classes down the side and the three functional classes across the top. In every cell that exists in the body, draw one example joint and name it. Leave the impossible cells empty and cross them out.

Then: Write beside the grid the one thing the structural classes are sorted by, and the one thing the functional classes are sorted by.
- ☐ **B TWO SORTING TREES**
Draw two trees. Tree one starts with the question, what holds the bones together, and branches to fibrous, cartilaginous, synovial, each ending in a small drawing of that joint. Tree two starts with the question, how much does it move, and branches to synarthrosis, amphiarthrosis, diarthrosis, each ending in a body example.

Then: Draw a line between the two trees for every branch on one tree that always lands on the same branch of the other.

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26

COMP

WEEK 6 · ARTICULATIONS AND JOINTS

Fibrous and cartilaginous joints

You should be able to: Describe the three fibrous joint types and the two cartilaginous joint types by structure, mobility, and example.

PICK ONE AND DO THAT ONE

- ☐ **A FIVE LABELED DRAWINGS**
Draw the five joints as simple cross sections: suture, syndesmosis, gomphosis, synchondrosis, symphysis. In each, draw what sits between the two bones and label it, dense fibrous tissue, a ligament, a periodontal ligament, hyaline cartilage, or fibrocartilage. Write the body example under each.

Then: Mark each drawing with how much it moves, and circle the one that is a tooth in its socket.
- ☐ **B SORTING TABLE**
Draw a table with the five joint types down the side, grouped into fibrous and cartilaginous. Across the top: what connects the bones, how much it moves, its functional class, and one example. Draw the connecting tissue in the first column.

Then: Put a box around the two rows where the connecting tissue is cartilage and write which kind of cartilage each one uses.

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27

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WEEK 6 · ARTICULATIONS AND JOINTS

Synovial joint structure and accessory parts

You should be able to: Identify the structures of a synovial joint, its accessory structures such as menisci, labra, bursae, and tendon sheaths, and its nerve and blood supply.

PICK ONE AND DO THAT ONE

☐ **A LABELED CROSS SECTION**

Draw a synovial joint cut through the middle. Label the articular cartilage on each bone, the joint cavity, the synovial membrane, the fibrous layer of the capsule, and the synovial fluid. Add a bursa outside the capsule, a tendon sheath around a tendon, and a meniscus inside the cavity, and label all three.

Then: Draw a nerve and a small artery reaching the capsule, and write which layer each one supplies.

☐ **B INSIDE-OUT HUB**

Put the joint cavity in the middle of the box. Spoke outward in layers: the fluid, the cartilage, the synovial membrane, the fibrous capsule, the ligaments. Then add a second ring of spokes for the accessory structures, meniscus, labrum, bursa, tendon sheath, each with a tiny drawing of where it sits.

Then: Write beside each accessory structure one joint in the body where you would find it.

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WEEK 6 · ARTICULATIONS AND JOINTS

Movements at synovial joints

You should be able to: Demonstrate and name the gliding, angular, rotational, and special movements permitted at synovial joints.

PICK ONE AND DO THAT ONE

- ☐ **A STICK FIGURE MOVEMENT SHEET**
Draw a stick figure for every movement: flexion and extension, abduction and adduction, circumduction, rotation, pronation and supination, dorsiflexion and plantar flexion, inversion and eversion, protraction and retraction, elevation and depression, opposition. Show the starting position and the moved position with an arrow between them, and label each.

Then: Group the figures into gliding, angular, rotational, and special, and draw a box around each group.
- ☐ **B PAIRED-OPPOSITES TABLE**
Draw a table with the movement pairs down the side and across the top: the joint where it happens, the plane it moves in, and a tiny stick-figure sketch. Fill the sketch column with drawings, and put the two members of each pair in one row so the opposites sit together.

Then: Circle the movements that have no opposite and write which joint each belongs to.

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WEEK 6 · ARTICULATIONS AND JOINTS

Six synovial joint types

You should be able to: Name the six synovial joint types with their axes of movement and give a body example of each.

PICK ONE AND DO THAT ONE

- ☐ **A SIX LABELED JOINT SHAPES**
Draw the six joint types as the shapes of their bone ends: plane, hinge, pivot, condylar, saddle, ball-and-socket. Draw the arrows for the way each one moves, and write the body example under each. Draw them in order from the least movement to the most.

Then: Mark each shape with how many axes it moves around, and draw the axes as lines through the joint.
- ☐ **B AXES SORT**
Draw three columns headed one axis, two axes, three axes. Put each of the six joint types in the right column with a small drawing of its shape and a body example. Add a fourth column for the one joint type that only glides.

Then: Beside each example joint, draw a tiny stick figure of the movement it allows.

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30

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WEEK 6 · ARTICULATIONS AND JOINTS

Shoulder and knee joint structure

You should be able to: Describe the articulating bones and supporting ligaments of the glenohumeral and tibiofemoral joints and identify the rotator cuff, menisci, and cruciate and collateral ligaments.

PICK ONE AND DO THAT ONE

- ☐ **A TWO LABELED JOINTS**
Draw the shoulder from the front: the glenoid cavity, the head of the humerus, the labrum, the capsule, and the four rotator cuff tendons wrapping the head. Then draw the knee from the front with the patella removed: the femoral condyles on the tibial plateau, both menisci, the two cruciate ligaments crossing, and the two collateral ligaments on the sides.

Then: On each joint, circle the structure that holds the joint together most, and write what the joint depends on when that structure is torn.
- ☐ **B TWO-JOINT COMPARISON TABLE**
Draw a table with shoulder and knee across the top. Rows: the bones that meet, the shape of the joint, what deepens the socket, the ligaments that support it, the cartilage pads inside it, the muscles or tendons that stabilize it, and the direction it most often gives way. Draw the shape row instead of describing it.

Then: Draw a small side view of the knee under the table and label the cruciate ligaments so their crossing is visible.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

DRAW IT

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